

Consumptive water use and irrigation performance of strawberries



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ABSTRACT

In south-west of Spain, the strawberry crop generates high economic value goods and high rate of employments. However, strawberry is cultivated in the vicinity of the Doñana National Park, a wetland with the maximum European environmental protection. Hence, minimize agricultural water use is of considerable interest in this area. Two experiments were conducted in order to determine strawberry (*Fragaria × ananassa*) irrigation requirements. The experiments were carried out with *Sabrina* and *Antilla* cultivars without deficit irrigation. Three irrigations amounts were applied with different water volumes. Drainage lysimeters were installed in order to measure crop evapotranspiration. In *Sabrina* trial, seasonal crop evapotranspiration ranged from 430 to 453 mm, whereas in *Antilla* it reached 352 mm. The crop coefficient reached maximum values of 1.1 and 0.8 in *Sabrina* and in *Antilla*, respectively. In *Sabrina* trial, irrigation efficiency of 81% was achieved when an irrigation volume of 5500 m³ ha⁻¹ was applied. In all *Sabrina* trial treatments, marketable fruit production exceeded 1000 g plant⁻¹ and crop productivities were above 74 t ha⁻¹, with no significant differences among treatments ($P < 0.05$). In *Antilla* trial, the maximum irrigation efficiency reached was 58%. The yield was around 750 g plant⁻¹, whereas land productivity was above 48 t ha⁻¹, with no significant differences between treatments. Water productivity was higher in *Sabrina* trial. The results suggest that following an irrigation schedule based on meteorological data and estimated crop coefficients, can result in significant water savings without losses in yield.

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1. Introduction

Spain is the main strawberry growing area in Europe and the fourth worldwide after the United States, Mexico and Turkey (FAOSTAT, 2014). An area of 7500 ha was dedicated to strawberry (*Fragaria × ananassa*) cultivation in the province of Huelva (southern Spain) in the 2012/2013 season. This figure represents a 10% increase compared to the average in the 2004–2007 period, when the average annual production was 300,000 metric tons (Consejería de Agricultura y Pesca, 2012). The strawberry crop is cultivated under plastic tunnels and generates high economic value goods and employment (around 55,000 jobs). Montesinos et al. (2011) described a high economic water productivity of 21.4 € m⁻³. However, the fact that it is cultivated in the vicinity of the Doñana National Park, the most important wetland in Europe, requires the reconciliation of environmental conservation and productive activity (Fig. 1). In fact, agricultural withdrawals for irrigation have

caused a general decline in groundwater and piezometric levels (Custodio et al., 2008). The Spanish Geological and Mining Institute (IGME) also highlights that during the last 30 years the average summer flows recorded in the main streams that flow into the Doñana marshes has suffered a 50% reduction (García Morillo et al., 2015a). Furthermore, the cation exchange capacity of these sandy soils is low and there can be significant leaching of crop nutrients from excess water (Albregts and Howard, 1984). Consequently, finding ways to reduce agricultural water use is of considerable interest in this area.

Despite the widespread use of drip irrigation in strawberry, there is still uncertainty about the precise amount of water needed. In previous works, a wide range of consumptive water use has been reported, between 300 mm (Trout and Gartung, 2004) and 797 mm (Strand, 2008). Likewise, crop yield ranged from 179 g plant⁻¹ (Rolbiecki et al., 2001) to 1704 g plant⁻¹ (Larson and Shaw, 1996). Moreover, in terms of land productivity, values ranging from 6 t ha⁻¹ (Stevens et al., 2011) to 83 t ha⁻¹ (Serrano et al., 1992) were found. Even in water productivity there is a wide range of values between 4.0 kg m⁻³ (Kirschbaum et al., 2003) and 14.67 kg m⁻³ (Serrano et al., 1992). In our study area there is also a similar situation with water applied between 4071 and 15,214 m³ ha⁻¹ (García

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Fig. 1. Location of the Doñana National Park and the study area.

Morillo et al., 2015a). In Huelva, strawberry farmers schedule their irrigations in order to obtain maximum production. They based on past experience, observation of weather conditions and visual plant indicators of stress. Therefore, over-irrigation easily occurs. This involves not only a problem of irrigation efficiency, but also of water pollution.

To improve their irrigation practices farmers need tools to calculate the strawberry irrigation requirements. In this sense, FAO developed a methodology based on crop coefficient (K_c) and reference evapotranspiration (ET_o) (Doorenbos and Pruitt, 1977). This methodology has been used worldwide to determine crop water requirements. FAO-56 Irrigation and Drainage Paper (Allen et al., 1998) manual suggests some general values of K_c for growing strawberries. However, there are differences between cultivars, length of growing season, weather conditions and farming systems. These differences raise questions about the applicability of these general values in any system. Likewise, a standardized method for estimating ET_o has been defined in the FAO-56 manual. Nevertheless, there is no standardized method under plastic tunnel conditions. This fact is important because strawberries are grown in plastic tunnels in most parts of the world.

In Europe they have been grown successfully since the early 1970s. In China, there are also thousands of hectares of strawberries grown over the winter in energy-saving plastic tunnels. However, most of the researches on strawberry irrigation have been carried out in California and Florida (USA), where open-field cultivation is the standard strawberry production method. McNiesh et al. (1985) estimated crop coefficients reporting values from 0.45 to 0.7 during

the summer production period. Clark et al. (1996), using water balance lysimeters in Florida, measured maximum K_c values of around 0.8 when the 60 cm-wide beds had nearly full cover. Grattan et al. (1998) measured strawberry ET_c using a Bowen ratio system in California. They estimated K_c as nearly equal to canopy cover, when there was 100% coverage on the bed. For the State of California, Hanson and Bendixen (2004) reported maximum K_c estimated values of 0.7, when the maximum coverage of the crop was no higher than 75%. However, these last authors did not measure ET_c . They estimated ET_c by first establishing the relationships between canopy coverage and days after planting (DAP). In this study, seasonal ET_c ranged from 310 to 396 mm. Jackson (1992) also reported maximum K_c values of 0.7 for final crop coverage of 75%. Finally, the FAO-56 Irrigation and Drainage Paper (Allen et al., 1998) lists basal crop coefficient values for strawberry as 0.15 early in the season, 0.85 midseason, and 0.7 late season.

In Spain almost all strawberries are grown under tunnels, but there are no studies that have attempted to estimate irrigation requirements in these conditions. Hence, since 2011, the Andalusian Institute of Agricultural Research and Training (IFAPA in Spanish) has conducted studies on drip irrigation of strawberries in Huelva. The aim is to help strawberry growers to obtain irrigation requirement values for their local crop system (Gavilán et al., 2014).

This study was conducted in Huelva coast to: (1) measure crop evapotranspiration of two commonly used strawberry cultivars using drainage lysimeters; (2) estimate their crop coefficients; (3)

evaluate their irrigation efficiencies, crop yields and water productivities.

2. Materials and methods

2.1. Field site description

Experiments were carried out in the 2012/2013 season, in Almonte (southern Spain), on two commercial strawberry (*Fragaria × ananassa*) farms with *Sabrina* (*Sab*) and *Antilla* (*Ant*) cultivars. *Sabrina* was the most planted cultivar in Huelva in the study season with 34.75% of the total cultivated area. On the other hand, *Antilla* is a classical cultivar in Huelva with a 5.05% of total cultivated area in this season. The farms are near the village of El Rocío (longitude 6°29' West, latitude 37°07' North, at an altitude of 75 m above sea level). The climate is Csa (Köppen–Geiger classification) with rainfall annual mean value of 467 mm. Average, maximum and minimum temperatures annual mean values are 17.8, 36.5 and 14.1 °C, respectively. The soils of the two studied areas are classified as sandy (USDA classification), with 90% sand and 10% clay for *Sabrina* trial, and 92% sand, 4% silt and 4% clay for *Antilla* trial.

Farm managers carried out all the necessary tasks for strawberry cultivation in the trials except drip irrigation. They contributed building the beds, maintaining their structure before planting and ensuring plant establishment after transplanting.

The strawberry plants were transplanted on October 9th and 15th 2012, with a planting density of 72,727 and 66,672 plants per ha in *Sabrina* and *Antilla*, respectively. The differences in crop systems for both cultivars are due to different farmer habits and not to agronomic constraints. The plastic tunnels were set up on November 10th and 19th, 2012, respectively. Hence, strawberries were grown outdoor until the second and the third week of November, respectively, and then they were covered. The berries were planted in trapezoidal raised beds measuring 0.60 m at the base, 0.50 m at the top, with a height of 0.50 m. Beds were 1.1 m apart in the first experiment (*Sab*), 1.2 m in the second (*Ant*), and were covered with black plastic mulch. Two rows of plants were placed along the bed with a subsurface drip irrigation tape in the center installed during bed construction. Plants were spaced 0.25 m apart. Each tunnel had six beds. *Sabrina* and *Antilla* fruit harvest started on January 9th and on January 20th, respectively. The experiments ended on June 6th (*Sab*) and on June 12th 2013 (*Ant*). Mature strawberry fruits were harvested twice a week from January to March and three times a week from March to June. In this study we only consider market category which is of great economic importance to growers.

A one-factor experimental design was used to compare three different irrigation regimes on two different cultivars. Both trials were arranged as a randomized block design, with three treatments of different irrigation volumes replicated four times. Irrigation was daily applied. The first treatment (T1) was set up to apply the crop water requirements, based on an approximation to ET_0 and estimated crop coefficients. Irrigation efficiencies of 80% (*Sab*) and 75% (*Ant*) to assure non-deficit irrigation were used. The lower value in *Antilla* was due to the fact that its soil was extremely sandy. It was done because the main goal of this study was to obtain crop water use values without deficit irrigation. T2 and T3 were designed to apply 25% and 50% more water respectively than T1. The experimental plot was a complete tunnel of $70 \times 6.6 \text{ m}^2$ (*Sab*) and $51 \times 7.20 \text{ m}^2$ (*Ant*). Each trial was carried out in 12 tunnels. Thus, the harvest area, in an almost flat site, was 5544 m^2 (*Sab*) and $4,406 \text{ m}^2$ (*Ant*). The tunnels were covered with 0.15 mm-thick thermal polyethylene plastic film. Using the entire tunnel as the experimental unit allowed the farmer to compare the results with the production and irrigation applied in the rest of the farm.

2.2. Meteorological conditions

Table 1 summarizes the measured meteorological values for each month of the growing season. Solar radiation is the main energy source inside the plastic tunnel. Plastic tunnel weather conditions varied from month to month. From October to February there is a light rise in evapotranspiration demand, whereas from March to June the demand is more exigent. The plastic tunnels were not whitened to reduce temperatures.

2.3. Irrigation scheduling

In both trials, water was supplied by drip irrigation using flow emitters spaced every 0.2 m, with a flow rate of $5 \text{ l m}^{-1} \text{ h}^{-1}$ at a pressure of 0.55 MPa. After planting, sprinkler irrigation was used until the plants were well established. This period included the open-field time described before. Thus, irrigation scheduling based on crop water requirements was completely applied during the covered phase. Irrigation was applied every day to replace ET_c and maintain non-limiting soil water content. Evapotranspiration was calculated by the FAO methodology (Doorenbos and Pruitt, 1977):

$$ET_c = ET_0 \times K_c \quad (1)$$

Before plastic tunnel were setup, ET_0 was estimated according to standardized Penman–Monteith equation proposed by Allen et al. (1998). However, there is no standardized method for calculating ET_0 inside greenhouse. After setting up the plastic tunnels, ET_0 was estimated inside them using the method proposed by Fernández et al. (2010) (Eqs. (2) and (3)). It was developed on the southeastern Mediterranean coast, at similar latitude than Doñana National Park.

$$\text{If } \text{DOY} < 220 \quad ET_0 = (0.288 + 0.0019 \times \text{DOY}) \times R_{\text{sin}} \quad (2)$$

$$\text{If } \text{DOY} > 220 \quad ET_0 = (1.339 - 0.0028 \times \text{DOY}) \times R_{\text{sin}} \quad (3)$$

where DOY is the day of year and R_{sin} is the daily solar radiation inside the tunnels (mm day^{-1}).

Thus, it is a practical way to determine ET_0 inside plastic tunnels. Moreover, its accuracy is similar to that of the Makkink FAO24 and Penman–Monteith FAO56 methods (Fernández et al., 2010; Gavilán et al., 2015). These equations only require greenhouse transmissivity (τ) and outdoor solar radiation (R_{sout}) data ($R_{\text{sin}} = R_{\text{sout}} \times \tau$). R_{sout} was estimated from forecasted temperatures supplied by the Spanish Meteorology National Agency (AEMET) using the Hargreaves equation (Hargreaves and Samani, 1982). Transmissivity depends on the material used for coverage, as well as the age and dirtiness of the plastic. Measurements of outdoor and indoor solar radiation were done during two previous growing seasons using two pyranometers. It was estimated that an average value of 0.75 transmissivity would be suitable for this type of plastic tunnels (Gavilán et al., 2015).

Plastic tunnel ET_0 values are usually lower than outdoor ET_0 values. The last ones were estimated using Penman–Monteith FAO-56 equation and lysimeter ET_0 measurements in the Mediterranean area. Indoor ET_0 values were approximately 21% lower than outdoor values. It was due to the fact that solar radiation and wind speed inside the tunnels were reduced (Gavilán et al., 2015).

Values of K_c recommended by Hanson and Bendixen (2004) were used in irrigation scheduling, except in trial of *Sabrina* at the end of the season (Table 2). An unexpected decrement in *Sabrina* drainage forced to use higher values of K_c to avoid deficit irrigation. Values of 0.8 from April 19th and 0.9 from May 1th to the end of the season were finally used.

Irrigation was the only source of water for the plants. During the first month after planting, and to ensure the root system establishment, overirrigation was applied. An irrigation time of 10 (*Sab*) and 15 (*Ant*) minutes per day was applied till January 27th. It was

Table 1

Average maximum (Max), minimum (Min), and mean (Avg) temperature (°C) and relative humidity (%) of the air, and solar radiation ($\text{MJ m}^{-2} \text{ day}^{-1}$) inside and outside greenhouse.

Month	Inside plastic tunnel							Outside plastic tunnel						
	Temperature (°C)			Relative humidity (%)			Solar radiation ($\text{MJ m}^{-2} \text{ day}^{-1}$)	Temperature (°C)			Relative humidity (%)			Solar radiation ($\text{MJ m}^{-2} \text{ day}^{-1}$)
	Max	Min	Avg	Max	Min	Avg		Max	Min	Avg	Max	Min	Avg	
Oct ^a								23.7	12.9	17.5	97.7	52.5	80.8	12.4
Nov ^b	20.7	7.1	12.6	95.1	41.8	78.8	6.9	19.2	10.3	14.2	99.6	62.3	87.8	8.4
Dec	21.9	6.8	12.1	98.5	51.4	86	6.2	17.2	6.1	10.9	100	67.8	92.5	7.9
Jan	21	6.7	11.9	97.9	50.8	83.2	7.4	17.1	5.8	10.8	99.1	58.5	87.1	9.5
Feb	21.1	5.2	11.3	96.2	41.4	76.7	10.5	16.8	4.4	10.1	97.3	45.9	77.9	13.1
Mar	21.9	9.8	14.3	98	55.4	83.8	11.1	18.2	9.3	13.3	97.9	59.1	83.8	12.8
Apr	25.8	10.5	16.7	96.9	38.8	73.6	17.4	22.9	10	15.9	96.7	40.9	73.9	21.2
May	27	10.9	17.9	95.5	34.9	69.1	19	24.7	10.3	17.3	94.1	34.2	67	23.6
Jun ^c	27	12	19.2	96.2	36.5	67.6	20.8	26.9	11.6	19.1	93	33.8	64.3	25.7

^a Measured values from October 10th to 31th.

^b Inside plastic tunnel measured values from November 10th to 30th.

^c Measured values from June 1st to 12th.

Table 2

15-day Hanson and Bendixen crop coefficient (K_{CH-B}) and *Sabrina* (K_{csche_Sab}) and *Antilla* (K_{csche_Ant}) scheduled crop coefficients.

Date	K_{CH-B} (mm d ⁻¹)	K_{csche_Sab} (mm d ⁻¹)	K_{csche_Ant} (mm d ⁻¹)
Oct 9–15 ^a	0.15	0.15	
Oct 16–31	0.17	0.17	0.17
Nov 1–15	0.24	0.24	0.24
Nov 16–30	0.34	0.34	0.34
Dec 1–15	0.45	0.45	0.45
Dec 16–31	0.55	0.55	0.55
Jan 1–15	0.61	0.61	0.61
Jan 16–31	0.65	0.65	0.65
Feb 1–15	0.67	0.67	0.67
Feb 16–28	0.68	0.68	0.68
Mar 1–15	0.69	0.69	0.69
Mar 16–31	0.69	0.69	0.69
Apr 1–15	0.69	0.69	0.69
Apr 16–30	0.69	0.78	0.69
May 1–15	0.7	0.9	0.7
May 16–31	0.7	0.9	0.7
1–6/12 Jun ^b	0.7	0.9	0.7

^a *Sabrina* was planted on October 9th and *Antilla* on October 15th.

^b *Sabrina* ended on June 6th and *Antilla* on June 12th.

done to assure good distribution uniformity (Hanson et al., 2003). In similar conditions, distribution uniformities lower than 80% were measured when irrigation time lasted less than 10 min (Lozano et al., 2014). In *Antilla* trial higher irrigation time of 15 min per day, instead of 10, was applied due to the extremely sandy soil conditions.

2.4. Crop evapotranspiration and crop coefficients

Drainage lysimeters, which have long been used to estimate evapotranspiration (Wright, 1982), were installed in each treatment to measure deep percolation. Six lysimeters, one by treatment for each cultivar, were located in the center of the plots to minimize edge effects. They were made of polyester, reinforced with fiberglass and with the dimensions 1.40 m × 0.30 m × 0.62 m. The lysimeters were buried with their upper edge at ground level and then the strawberry beds were rebuilt over them. In each lysimeter, 11 plants were planted at a similar density than the rest of the bed. Each of them captures drainage from 7 emitters. Lysimeters were installed after beds were formed. After installation beds were rebuilt by hand. Readings of the drainage lysimeters were taken daily. ET_c values were calculated using the water balance method (Allen et al., 1998). These values were used to obtain

K_c for strawberries in a 15-day calculation period. ET_0 was estimated as described in Section 2.3 using measured meteorological data obtained from two weather stations (outdoor and indoor), belonging to the Meteorological Information Network of Andalusia (Gavilán et al., 2008). Weather stations are equipped with sensors for measuring temperature and relative air humidity (HMP 45C probe, Vaisala), solar radiation (pyranometer CM3, Kipp & Zonen) and wind speed and direction -only the outdoor- (Young wind monitor), all of them controlled by a data logger CR10X (Campbell Scientific).

As pointed out by Hanson and Bendixen (2004), crop coefficients expressed as a function of canopy coverage are considered more universal than those expressed on a time basis. The percent of ground surface covered by vegetation (f_c) was determined based on digital photography. Photographs over the lysimeter were taken in each treatment three times along the irrigation season (beginning, intermediate and ending). This was done at solar noon vertically from an approximate height of 2 m above the top of the bed. Image processing was carried out with the help of the ARGIS 10.2 computer program. First, the vegetation (V) and soil (S) were delineated. Next, the ground cover was calculated as the $V/(V + S)$ ratio as percentage.

However, growers are not likely to use crop coefficients based on canopy coverage. They consider that it is difficult and time consuming to estimate canopy coverage. Hence, growers prefer to use crop coefficients expressed on a time basis, such as DOY, DAP or growing degree-days (GDD) since planting. The GDD, involve the sum of daily mean temperature above a given base temperature. Cumulative growing degree-days for strawberry were calculated as described in Snyder et al. (1999). A base temperature from 0 to 5 °C was generally adequate for all strawberry cultivars, although a base temperature of 4 °C was often reported to be the most satisfactory for strawberry (Strik and Proctor, 1988).

The amount of water applied in each treatment was measured using flow meters. Irrigation was controlled with solenoid valves connected to an irrigation controller. They were installed at the head of each treatment, and thus they provided data on the volume applied to the group of four tunnels. In addition, flow meters were installed at the beginning of the drip lines in the beds where the lysimeters were located, in order to measure the amount of water applied to each lysimeter.

2.5. Irrigation performance

Four performance indicators were selected to evaluate water management and productivity of the treatments of this study:

- Field application efficiency (%): ratio of the irrigation water required for crop to the total volume furnished to the field.
- Water productivity (kg m^{-3}): ratio between the actual crop yield achieved and the water use.
- Land productivity (t ha^{-1}): ratio of the strawberry production to productive surface.
- Yield (g plant^{-1}): defined as the strawberry production by plant.

Field application efficiency is not only an irrigation efficiency indicator but also an indicator of fertilizer efficiency. Water drainage carries nutrient solution supplied by fertigation, causing so-called diffuse pollution. In this study, the use of lysimeters allows field application efficiency to be measured, which is more reliable than estimating it.

Note that production reported on this study refers to the whole plastic tunnel as a plot, whereas in the majority of previous works, fruit was harvested from a little plot (Guimerà et al., 1995; Hoppula and Salo, 2007) or from the center rows of each plot (Stevens et al., 2011).

2.6. Statistical analysis

Yield data were statistically analyzed using analysis of variance (ANOVA), according with the experimental design in which the experiment was carried out. LSD test was used to identify treatments that were statistically different at a level of significance of 0.05. Software Statistix Sx.9 was used in the analysis.

3. Results

3.1. Evapotranspiration and crop coefficients

The estimated ET_0 over the course of the growing season was 523 mm. Rainfall recorded at outdoor meteorological station during this period was 580 mm. There was no precipitation before the tunnels were erected. In Table 3 lysimeter water use data are expressed as the equivalent depth over a production area that includes beds and row middles. The cumulative average seasonal ET_c measured with lysimeters was 441 mm in *Sabrina*, with figures ranging from 430 mm (T3) to 453 mm (T1), representing around 85% of estimated ET_0 . These differences can be considered acceptable given the constraints associated with the use of drainage lysimeters. In the *Antilla* trial, the ET_c measurements were 271 mm (T1), 304 mm (T2) and 352 mm (T3). T1 and T2 lysimeters had significant lower values than T3. Plants in lysimeters T1 and T2 were smaller than those grown in the rest of the bed. A compaction problem of the beds above the T1 and T2 lysimeter could happen since plants sizes outside the lysimeters and yields were similar in all treatments. The bed was rebuilt by hand after lysimeter installation. This could lead to a lower bulk density and a higher hydraulic conductivity in relation to the rest of the bed. In this extremely sandy soil, this might have contributed to a shorter water available time in the narrow root zone. This effect was especially critical in the first stage of the season and it might have caused crop water stress in the lysimeter plants. Therefore, an ET_c value of 352 mm, representing around 67% of ET_0 , was considered appropriate. This value was then used to estimate K_c and calculate irrigation performance indicators (Table 4).

Measured crop water use increased as the season moved forward (Fig. 2). The maximum crop evapotranspiration month was June, in which average daily values of 5.5 and 3.6 mm d^{-1} were

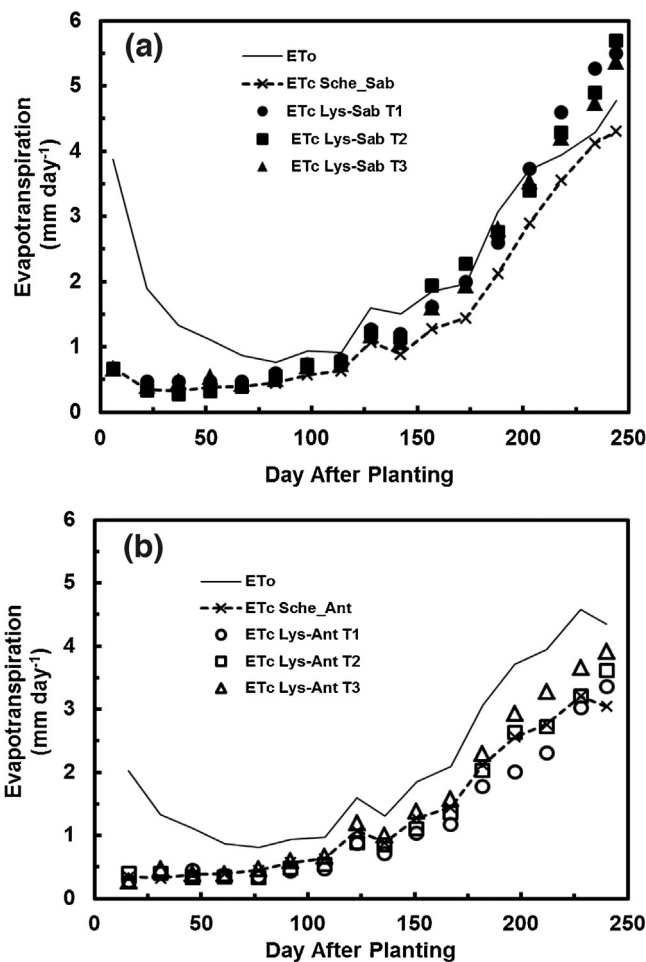


Fig. 2. Evolution of 15-day reference evapotranspiration (ET_0), lysimeter crop evapotranspiration (ET_{clys}) and evapotranspiration used in irrigation scheduling (ET_{cSche}) for (a) *Sabrina* and (b) *Antilla* along the season.

reached by *Sabrina* and *Antilla*, respectively. In November, values of 0.4 mm d^{-1} (*Sab*) and 0.3 mm d^{-1} (*Ant*) were reached. The average daily crop ET_c over the course of the season was 1.8 mm d^{-1} for *Sabrina* and 1.5 mm d^{-1} for *Antilla*. Along the season, three stages were identified according to the slope of Fig. 2. In *Sabrina* trial, up to about 114 days after planting the daily ET_c was less than 1 mm d^{-1} . Later, from 114 to 173 days after planting, the daily ET_c was between 1 and 2 mm d^{-1} . Finally, from 173 DAP onwards, ET_c and ET_0 increased linearly with maximum values around 5.5 and 4.8 mm d^{-1} , respectively (Fig. 2a). Three stages of evapotranspiration were also identified in *Antilla*. The daily ET_c was under 1 mm d^{-1} , up to about 128 DAP. ET_c was between 1 and 2 mm d^{-1} from 128 to 167 DAP. After 167 DAP, ET_c increased linearly with maximum values around 3.6 mm d^{-1} (Fig. 2b).

A time-based K_c function of the DAP was developed. In the case of *Sabrina*, the initial crop coefficient was around 0.2 in early November (Fig. 3a). The crop coefficient increased rapidly with DAP up to about January 1st. A rapid canopy-growth stage occurred from 40 to 100 DAP. This growth then continued more slowly and the ratio increased progressively until mid-February, reaching values close to 0.8. From that point on, the crop coefficient slowly increased until it peaked at 1.1 in mid-May (225 DAP). In the *Antilla* trial, the initial crop coefficient was around 0.15 in early November (Fig. 3b). The crop coefficient increased rapidly with DAP up to about 1st January. A rapid canopy-growth stage occurred during the first 120 DAP. This growth then continued more slowly and the

Table 3T1, T2 and T3 irrigation supply and lysimeter crop evapotranspiration (ET_{clys}) for a) *Sabrina* and b) *Antilla*.

Period	<i>Sabrina</i>				<i>Antilla</i>			
	ET_{clys}^c (mm)	Irrigation ($m^3 ha^{-1}$)			ET_{clys}^d (mm)	Irrigation ($m^3 ha^{-1}$)		
		T1	T2	T3		T1	T2	T3
Oct 10–15 ^a	4.0	255	276	266				
Oct 16–31 ^a	6.0	258	293	304	4.0	60	60	61
Nov 1–15	6.0	111	139	164	7.0	176	191	235
Nov 16–30	6.6	105	128	175	5.8	184	207	272
Dec 1–15	6.4	160	181	189	5.9	198	191	250
Dec 16–31	8.3	190	201	200	7.0	250	226	258
Jan 1–15	10.8	185	225	254	9.0	271	249	266
Jan 16–31	11.5	178	231	275	10.0	239	289	329
Feb 1–15	18.3	259	322	387	18.0	302	369	416
Feb 16–28	17.0	234	291	341	15.0	314	369	437
Mar 1–15	25.7	286	368	420	20.8	365	453	550
Mar 16–31	31.0	360	464	548	23.7	494	587	704
Apr 1–15	40.8	412	536	608	34.5	548	657	774
Apr 16–30	53.3	575	696	813	43.9	671	774	958
May 1–15	65.4	734	919	1065	49.2	714	817	993
May 16–31	74.5	790	1006	1199	55.0	765	900	1086
1–6/12 Jun ^b	55.2	550	601	749	43.0	620	704	871
Total	440.8	5642	6876	7957	351.8	6171	7042	8460

^a *Sabrina* was planted on October 9th and *Antilla* on October 15th.^b *Sabrina* ended on June 6th and *Antilla* on June 12th.^c ET_{clys} average value of T1, T2 and T3.^d ET_{clys} value of T3.

ratio increased progressively until the end of the season, reaching values close to 0.8.

To facilitate extrapolation of the results to other areas, the lysimeter crop coefficient values for both *Sabrina* and *Antilla* were compared to the evolution of ground cover. As shown in Fig. 4, K_c was a linear function of ground cover with a high coefficient of determination. This linear regression is similar to the relationship found by Grattan et al. (1998) for strawberries grown in San Diego County near the Oceanside CIMIS station. Crop coefficients values in function of canopy cover for strawberry were also similar to those values obtained by Allen and Pereira (2009).

According to generalize the found lysimeter K_c for different environmental conditions, a quadratic regression analysis was carried out. K_c was taken as the dependent variable and GDD as independent variable (Fig. 5). The K_c -model obtained for *Sabrina* cultivar was $K_c = -0.0000001GDD^2 + 0.0007GDD + 0.0109$ with a determination coefficient $R^2 = 0.95$. The model obtained for *Antilla* cultivar was $K_c = -0.0000001GDD^2 + 0.0007GDD + 0.0644$ with a determination coefficient $R^2 = 0.94$.

3.2. Irrigation applied

The volumes delivered to each of the treatments of *Sabrina* trial were 5,642, 6,876 and 7,957 $m^3 ha^{-1}$ for T1, T2 and T3, respectively. Fertigation was applied at the same concentration for all treatments. Similar fertilizer concentrations were measured in both irrigation and lysimeter water. For T2 and T3 this represented, respectively, 23 and 42% more water and fertilizer than in T1.

In treatment T1, November was the month when less water was applied ($7.2 m^3 ha^{-1} day^{-1}$), while the maximum value of water applied was registered in June ($51.8 m^3 ha^{-1} day^{-1}$). The irrigation target was achieved since only 2% more water than expected was applied in treatment T1. In *Antilla* experiment, the volumes delivered were 6171, 7040 and 8460 $m^3 ha^{-1}$ for T1, T2 and T3, respectively. For T2 and T3 this represented 14 and 37% more water and fertilizer than in T1. In treatment T1, November was the month when the least amount of water was applied ($12 m^3 ha^{-1} day^{-1}$), while maximum value was registered in May ($52 m^3 ha^{-1} day^{-1}$). The irrigation target was not achieved since more water than was expected was applied in treatment T1.

3.3. Yield and water productivity

In the *Sabrina* experiment, seasonal market yields were over 1000 g plant⁻¹ in all treatments (Table 4). In the *Antilla* experiment, crop yields were around 750 g plant⁻¹ (Table 4). No significant differences between them were found in both trials. Hence, there is a negligible effect on yield in the three treatments with these water volumes applied. These results show that treatments cannot be considered in deficit.

Due to higher plant density used in *Sabrina*, land productivity in *Sabrina* was around 56% higher than in *Antilla* (Table 4). However, in terms of yield (g plant⁻¹) *Sabrina* was around 37% higher than *Antilla*. Therefore, land productivity should be a more interesting performance indicator than crop yield for farmers. However, the

Table 4Irrigation performance indicators: field application efficiency (%), yield (g plant⁻¹), land productivity (t ha⁻¹) and water productivity (kg m⁻³).

	<i>Sabrina</i> cv.			<i>Antilla</i> cv.		
	T1	T2	T3	T1	T2	T3
Field application efficiency (%)	81	65	54	58	50	42
Yield ^a (g plant ⁻¹)	1027*	1028*	1084*	731*	754*	745*
Land productivity (t ha ⁻¹)	74.7	74.8	78.9	48.7	50.3	49.7
Water productivity (Kg m ⁻³)	16.5	16.8	18.3	13.8	14.3	14.1

^a Means followed by the same asterisk (*) in each row do not differ significantly ($P < 0.05$).

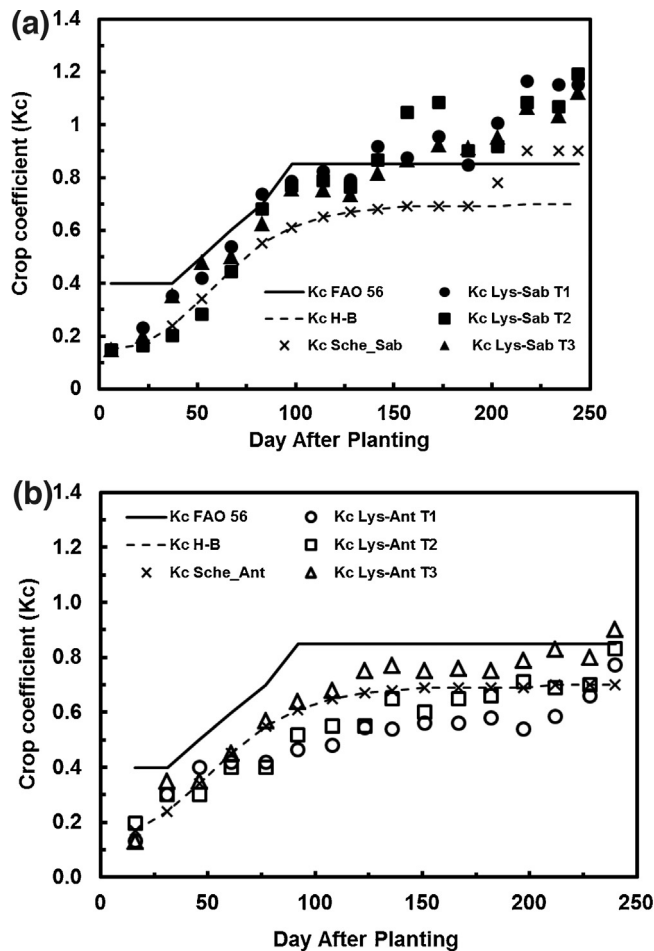


Fig. 3. Evolution of 15-day K_c lysimeter (K_{clys}), K_c used in irrigation scheduling (K_{cSche}) and estimated K_c according to Hanson and Bendixen (K_{cHB}) and FAO-56 (K_{cFAO56}) for (a) *Sabrina* and (b) *Antilla* along the season.

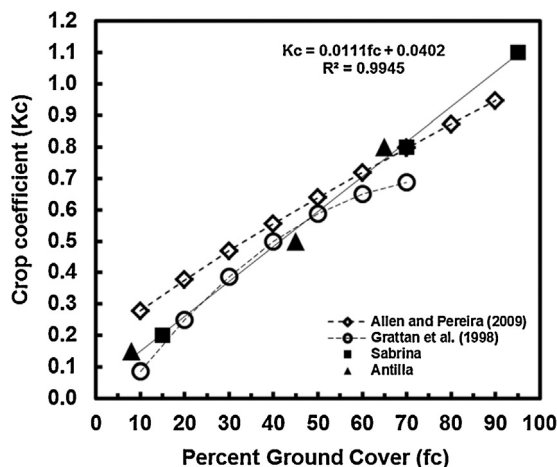


Fig. 4. Crop coefficient (K_c) versus percent ground cover (fc) reported by Grattan et al. (1998) and Allen and Pereira (2009) and measured values in *Sabrina* and *Antilla*. The solid line represents a linear regression using data from *Sabrina* and *Antilla*.

majority of the farmers of this area use yield as the main productivity indicator.

Finally, water productivities for *Sabrina* in each treatment were 16.5, 16.8 and 18.3 kg m⁻³ for T1, T2 and T3, respectively. For *Antilla*, water productivities were 13.8, 14.3 and 14.1 kg m⁻³ for T1, T2 and T3, respectively.

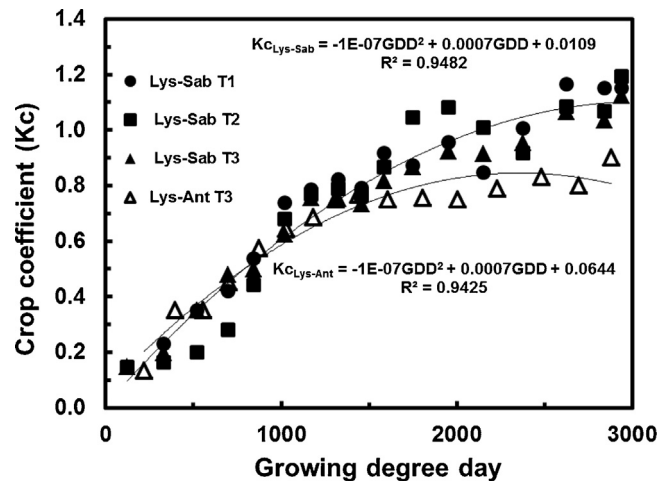


Fig. 5. Crop coefficient (K_c) as a function of growing degree day (GDD) for both strawberry cultivars, *Sabrina* and *Antilla*.

4. Discussion

4.1. Strawberry water use and crop coefficient

A wide range of consumptive water use, ranging between 300 and 787 mm, was reported in previous works (Serrano et al., 1992; Hanson and Bendixen, 2004; Trout and Gartung, 2004; Strand, 2008). Seasonal ET_c values for *Sabrina* in our study was similar to estimated values reported by García Morillo et al. (2015b) also in Huelva, ranging from 460 to 487 mm. The average maximum crop coefficients value of 0.7 measured in our *Antilla* trial was similar to those reported in some previous works (Hanson and Bendixen, 2004; Trout and Gartung, 2004). Lower maximum values of 0.6 were also described by Roudeillac and Veschambre (1987) and Krüger et al. (1999). Allen et al. (1998) recommended maximum values of 0.85, higher than in *Antilla*. However, maximum crop coefficients of 1.1, as were measured in *Sabrina* trial, were not found in previous works.

4.2. Strawberry water applied

Growers of the study area perceive that strawberry is very sensitive to water stress. Average irrigation amount in the farm around the *Sabrina* and *Antilla* trial was 8,235 m³ ha⁻¹ and 10,000 m³ ha⁻¹, respectively. Guimerà et al. (1995), Kirschbaum et al. (2003) and Kirnak et al. (2003) reported irrigation amounts around 7000 m³ ha⁻¹ for non-deficit irrigation trials. In the first work, the duration of the irrigation season was from November to July. One month more than in our study. Kirschbaum et al. (2003) trial took place from July to December. This trial was done in the southern hemisphere (Argentina). Kirnak et al. (2003) was a two-phase trial from August to November and from March to May. The applied volumes in these studies were higher than irrigation volumes applied in T1 (Table 2). Berbel et al. (2011) described an average dose of 6,000 m³ ha⁻¹ in Huelva, similar to those obtained in our study. However, other authors reported lower values. Yuan et al. (2004) applied 3,800 m³ ha⁻¹ from November to mid-June. This period is equal to our study period. Pires et al. (2006) applied 4,740 m³ ha⁻¹ from April to December. This work took place in Atibaia, Brasil, in the southern hemisphere.

In line with Yuan et al. (2004), stages in the irrigation season were defined. The last one, from March to June, is the most demanding period. These authors found that the amount of irrigation water in an experiment with 80% of irrigation efficiency was 3.43 mm day⁻¹. This was very close to the figure of

3.53 mm day⁻¹ obtained in *Sabrina* trial with similar irrigation efficiency. Kirschbaum et al. (2003) considered that irrigation frequency did not affect yield significantly. However, in sandy soils too low irrigation frequency can lead to lower yields (Hoppula and Salo, 2007; Pires et al., 2006; Serrano et al., 1992). It is possible that high irrigation frequency (up to six irrigations per day in days with high evaporative demand) used in this study played an important role in the results.

5. Conclusions

Drainage lysimeters has been designed and installed to determine consumptive water use and crop coefficients of strawberries under plastic tunnel. Crop coefficient values higher than those described previously in the literature were obtained for *Sabrina* and *Antilla* cultivars. The results show the need to adjust K_c to cultivar, local weather conditions and growing practices for an appropriate irrigation scheduling. The use of irrigation scheduling based on a time-based K_c function of the DAP or GDD, could result in significant savings in water and fertilizer without negatively affecting crop yield. The use of GDD may improve the estimate by removing season variations in crop development. Therefore, the information developed in this study allows farmers to calculate crop water requirements and to set irrigation scheduling. It could also significantly reduce environmental impact in terms of diffuse pollution and improve water productivity in strawberry. This is especially relevant in areas such as Doñana with limited water resources and a high environmental protection.

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